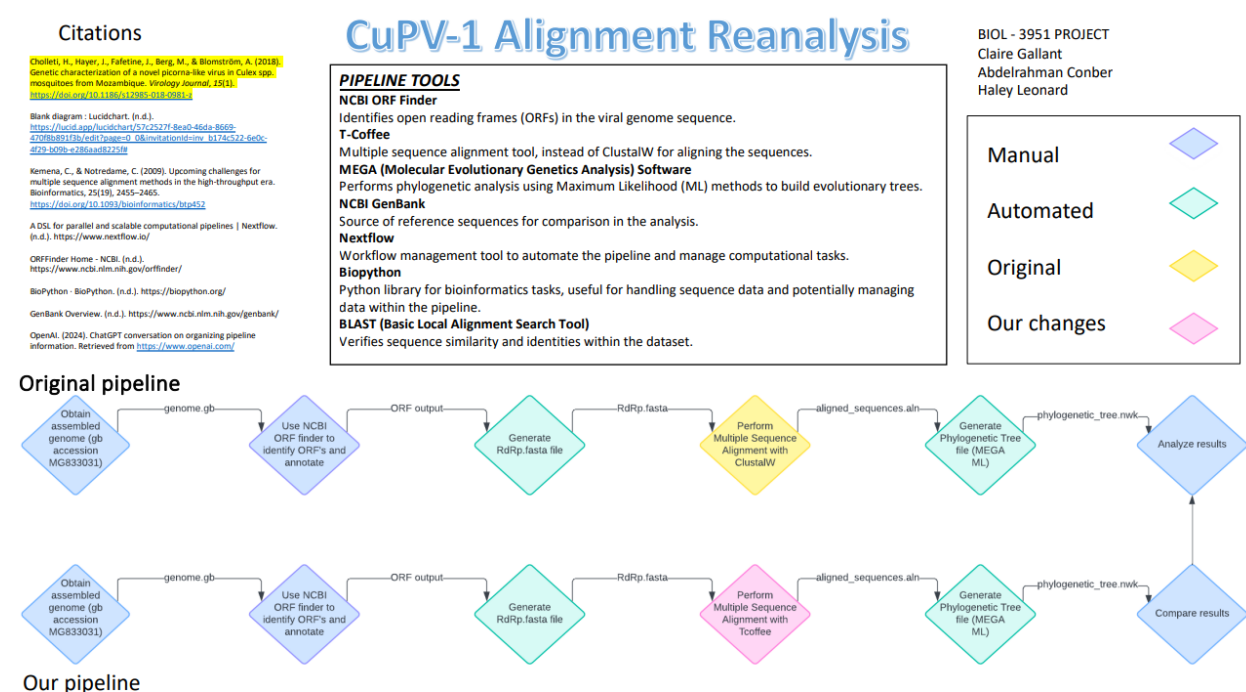


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1- What's your project topic?

Here is a 1-page plan (also our presentation of our project):



We also questioned whether a 1-page plan means submitting the flowchart or an actual full 1-page plan of the project. So, we decided to do both. If the pictures suffice, that's great; if not, here is the detailed version and plan of the project:

Overview:

In this project, our group is aiming to reanalyze the alignment of the **CuPV-1 Virus**, a picorna-like virus found in **Culex** mosquitos. It was initially characterized through genome sequencing. In the original analysis, tools like **NCBI ORF Finder**, **ClustalW**, and **MEGA** were used to identify ORFs, align the sequence, and generate a phylogenetic tree. This process was crucial in providing valuable information. However, there are always ways to improve and enhance the workflow through automating pipelines, using more advanced alignment tools, and improving computational efficiency.

Goals:

1. Pipeline optimization and automation:

- We are hoping to implement **Nextflow** (a workflow management tool) to automate the pipeline.

2. Tool substitution for potential improvement:

- Replace **ClustalW** with **T-Coffee** for multiple sequence alignments, to potentially enhance alignment quality.

3. Comparison of pipelines:

- Perform comparative analysis between the original pipeline and our modified pipeline (using **T-Coffee**) to evaluate the effect of the change on the results and phylogenetic tree construction.

Methods:

1. Acquiring the data:

- We will be obtaining the viral genome (Accession number: **MG8330331**) from **NCBI GenBank** for reanalysis.

2. ORF identification:

- Using **NCBI ORF Finder**, we will identify and annotate the ORFs.

3. Sequence alignment (the modification):

- The original pipeline uses **ClustalW** for sequence alignment.
- Our modified pipeline will use **T-Coffee** as a replacement for **ClustalW** to achieve improved accuracy and alignment quality.

4. Constructing the phylogenetic tree/file:

- We will continue using **MEGA** to generate the phylogenetic tree.

5. Comparison and reanalysis:

- Both the old and modified pipelines will be used to compare and evaluate differences in the alignment, highlighting any improvements from the substitution of new (possibly better) tools.

Expected/hopeful outcomes:

1. Pipeline efficiency:

- Automating the pipeline using **Nextflow** is expected to reduce the time and effort required for manual execution, leading to a more efficient workflow.

2. Improved alignment accuracy:

- We expect that using **T-Coffee** instead of **ClustalW** will provide a more modern and reliable sequence alignment, which will in turn improve the overall accuracy of the phylogenetic tree when constructed.

3. Enhancing reproducibility:

- In bioinformatics, reproducible research is crucial. By automating the pipeline using **Nextflow**, we hope to achieve improved reproducibility, making it easier to apply the workflow to other genomes and similar analyses in the future.

Challenges and difficulties:

1. Tool compatibility:

- Ensuring that input/output formats between different bioinformatics tools are compatible may cause challenges. We hope that using **Nextflow** and **Biopython** will help us achieve a smoother workflow and automate the pipeline, addressing this obstacle.

Conclusion:

- Our group project aims to refine and optimize the **CuPV-1 viral alignment pipeline** by using modern tools and methods like **T-Coffee** and workflow systems like **Nextflow**. This will allow us to better understand the effects these tools produce when comparing our analysis.